

BIOMG Study Group Week 2

With Lani and Emily

To Knows Lecture 3

Learning Outcomes from Lecture 3:

- Describe the distinct functional roles of hemoglobin and myoglobin.
- Define prosthetic group, holoprotein, apoprotein.
- Describe the essential role that the myoglobin protein plays in heme group binding to oxygen.

Hemoglobin vs. Myoglobin: Transport vs. Storage

Hemoglobin (Hb): The Oxygen Transporter

- Location: Maintained inside red blood cells traveling through capillaries.
- Structure: A **tetramer** composed of **4 polypeptide chains**, binding up to **4 O₂** molecules.
- Primary Role: Provides the **transport mechanism** to move oxygen through the bloodstream to distant tissues.
- Functional Control: Sensitive to physiological control signals such as **low pH**, which alters ion pair bonds and **triggers the release of bound O₂** into capillaries, where it can diffuse into nearby cells

Myoglobin (Mb): The Oxygen Reservoir

- Location: Resides in the **cytosol of muscle** cells.
- Structure: A **monomer** composed of a **single polypeptide chain** (153 amino acids, MW ~17,000) containing a **non-amino acid heme prosthetic group**.
- Primary Role: Functions as an intracellular **storage form for oxygen** in tissues. Mb significantly increases the amount of O₂ stored in cells (by about 333-fold!)
- Functional Control: Binds O₂ once it enters the cytosol and delivers O₂ to the mitochondria.

Similarity: Both hemoglobin and myoglobin have the same globin fold (an 8-helix, compact structure that binds heme using two His residues, a proximal His that binds covalently to the heme and a distal His that stabilizes bound O₂).

Protein Components: Prosthetic Groups, Apoproteins, & Holoproteins

Key Definitions

- **Prosthetic Group:** A non-amino acid structure bound within a protein
- **Apoprotein:** The protein polypeptide chain without its prosthetic group attached
- **Holoprotein:** The entire, complete protein assembled with its prosthetic group

Application: The Myoglobin Complex

- Prosthetic Group = Heme: A non-amino acid ring structure with an Fe^{2+} iron center that binds molecular oxygen (O_2)
- Apoprotein = Myoglobin Polypeptide: The 153-amino-acid chain before the heme group is incorporated
- Holoprotein = Functional Myoglobin: The intact complex where a proximal Histidine residue covalently links the polypeptide to the heme group

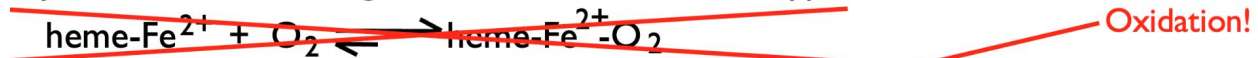
Why Heme Needs Myoglobin: Enabling Reversible O₂ Binding

The Problem: Free Heme Cannot Reversibly Bind O₂

- Redox Oxidation: Without the myoglobin protein, free heme in water undergoes a redox reaction that converts ferrous iron (Fe²⁺) to ferric iron (Fe³⁺)
- Reaction vs. Binding: Oxygen chemically reacts with free heme rather than binding to it reversibly
- Heme Stacking: Destructive chemical reactions occur because unshielded hemes stack together in aqueous solution

By itself, in water without protein, heme does not bind O₂ !

Just below is the binding reaction, but this does not happen without Mb! Instead of binding:

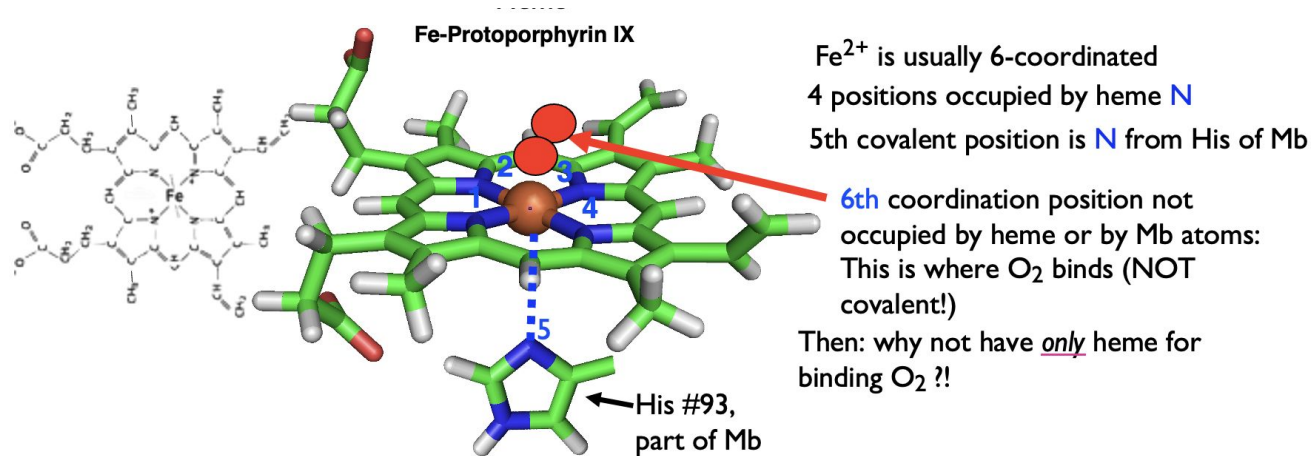


Oxidation!

Why Heme Needs Myoglobin (cont.)

How Myoglobin Anchors the Heme Group

- 6-Coordination Geometry: Heme Fe^{2+} is 6-coordinated: 4 positions are bound by heme nitrogens, the 5th position is covalently linked to a proximal Histidine (His) residue on the polypeptide chain, and the 6th site binds O_2
- Structural Stability: The covalent proximal His bond and a tight-fitting binding pocket hold the heme securely inside myoglobin



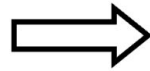
Why Heme Needs Myoglobin (cont.)

Essential Roles of the Myoglobin Polypeptide

- Prevents Heme Stacking: Holds individual hemes apart from one another so they cannot stack and react chemically
- Maintains Accessibility: Keeps Fe^{2+} close to the aqueous environment where oxygen is dissolved without permitting irreversible redox damage
- Enhances Storage Capacity: Allows intracellular oxygen levels to reach ~ 100 mM, compared to the low ~ 0.3 mM solubility of free O_2 in water

Function of Mb polypeptide (apoprotein):

a. Prevent hemes from touching each other, because above reaction involves formation of a 4-heme complex!



Mb holoprotein: tight heme (and O_2) binding without chemical reaction

b. Create 100s of fold higher $[\text{O}_2]$ because of high solubility of MbO_2

Why Heme Needs Myoglobin (cont.)

Explain why we don't simply use free heme to store O₂ in the body (i.e., why is myoglobin needed?)

- There are two reasons the myoglobin protein is needed.
 - First, the free heme can form tetramers that react with O₂, oxidizing the heme iron into Fe³⁺ that binds to H₂O, not O₂
 - Second, the myoglobin effectively increases the concentration of stored O₂ by more than two orders of magnitude, enabling the cell to have plenty of O₂ available when needed.

To Knows Lecture 4

Lecture 4 Learning Outcomes:

- Define isozyme/isoenzyme/isoform, recognize examples.
- Define “primary structure” and explain why it is important and ways it can be used.
- Recognize short sequence motifs.
- Define homolog, paralog, and ortholog.
- Understand why comparison of primary structures of orthologs provides evolutionary relationships among organisms.
- Describe bioinformatics and understand what AlphaFold does.
- List the three main ways the atomic-level 3D structure of proteins can be determined.
- Recognize names of scientists who played key roles in founding protein X-ray crystallography.

Isozymes: Definition, Examples, & Physiological Function

Key Definitions

- **Isozymes (Isoenzymes / Isoforms):** Enzymes from the same organism that are structurally (amino acid sequence and 3D fold) and catalytically similar, but **not identical**.
- **Structural Variation:** Isozymes share similar 3D structures but have slightly different amino acid sequences.
- **Functional Differences:** Sequence variations alter their substrate binding affinity, reaction rates, and regulatory mechanisms across different tissues.

Canonical Example: The Hexokinase (HK) Family

- **Kinase Function:** Kinases are enzymes that catalyze the formation of phosphate esters from nucleotide triphosphates.
- **Reaction:** Hexokinase catalyzes the reaction $\text{hexose} + \text{ATP} \rightarrow \text{hexose-PO}_4^{2-} + \text{ADP}$

Isozymes: Definition, Examples, & Physiological Function

- **The 4 Mammalian HK Isozymes:** (example - no need to memorize, but understand conceptually)
 - **HK I:** Expressed in **brain** and **red blood cells**.
 - **HK II:** Expressed in **muscle**.
 - **HK III:** Expressed in **lungs, liver, and spleen**.
 - **HK IV (Glucokinase):** Expressed specifically in the **liver**.

Physiological Significance: Tissue-Specific Regulation

- **Weak Glucose Affinity:** **HK IV** in the liver binds glucose much more weakly than HK I, II, or III.
- **Glucose Sparing:** This low affinity ensures the **liver does not compete with other tissues** for glucose under baseline conditions.
- **Blood Glucose Clearance:** When blood glucose levels become abundant (above 5 mM), HK IV binds glucose significantly, removing it from blood to store as glycogen.

Primary Structure: Definition, Importance, & Applications

Key Definition & Experimental Accessibility

- **Primary Structure:**
 - The specific linear sequence of amino acids in a polypeptide chain

Importance: The Blueprint for Structure & Function

- **Determines 3D Fold:**
 - The linear amino acid sequence contains the fundamental information required for a protein to fold into its native 3D structure
- **Homology & Evolutionary Tracking:**
 - Comparing sequences reveals homologs—either orthologs (from different organisms) or paralogs (from the same organism)—allowing researchers to infer shared ancestry and function
 - Because sequence data is fast to obtain, biochemists use it to predict 3D structures (homology models), deduce catalytic functions, locate organelle/regulation motifs, and trace evolutionary relationships

Sequence Homology, Ortholog Evolution, & Functional Motifs

Defining Related Sequences (Homologs)

- **Homologs:** Proteins with related amino acid sequences, typically sharing related structure and function
- **Paralogs:** Homologous proteins found within the same organism
- **Orthologs:** Homologous proteins found across different organisms

Orthologs & Evolutionary Relationships

- **Universal Proteins:** Certain essential proteins—such as Cytochrome c in the electron transport chain—are required across the spectrum of life, including plants, animals, and single-cell organisms
- **Primary Structure Comparison:** Comparing the linear amino acid sequences of orthologs across species allows biochemists to map how life forms are evolutionarily interconnected
- **Short sequence motifs:** serve as specific cellular zip codes (nucleus, ER, peroxisome) or biochemical switches (kinase phosphorylation, zinc-finger DNA binding)

Bioinformatics & AlphaFold

Can the amino acid sequence of a protein enable prediction of that protein's 3D structure?

Bioinformatics & AlphaFold

- **Bioinformatics:** A field of science dedicated to analyzing biological data and researching protein sequence analysis, genome annotation, comparative genomics, systems biology, and cancer mutations
- **AlphaFold (DeepMind):** A computational tool combining bioinformatics and physical data to predict native 3D protein structures (including sidechain positions) directly from linear amino acid sequences

3D Structure Determination

The 3 Main Experimental Methods for 3D Atomic Structure

1. X-Ray Diffraction of Protein Crystals

- **Method:** Analyzes X-ray scattering through crystallized protein samples
- **Impact:** Solved ~85% of all known protein 3D structures (the primary bottleneck is growing high-quality crystals)

2. Multi-Dimensional NMR (Nuclear Magnetic Resonance)

- **Method:** Analyzes proteins dissolved in aqueous buffer solution
- **Impact:** Solved ~15% of known structures (avoids crystallization, but requires larger quantities of protein)

3. Cryo-Electron Microscopy (Cryo-EM)

- **Method:** Uses high-sensitivity, direct electron-counting cameras and advanced image processing software
- **Impact:** A major modern breakthrough enabling atomic-level structure determination without requiring crystallization

Typically, are the structures of a protein determined by x-ray diffraction and by NMR in solution very different from each other? Why?

No, in most cases they are the same or very similar. This is because protein crystals are typically composed of ~50% water by volume, so the protein is in an environment that is similar to aqueous solution.

To Knows Lecture 5

Lecture 5 Learning Outcomes:

Be able to describe ...

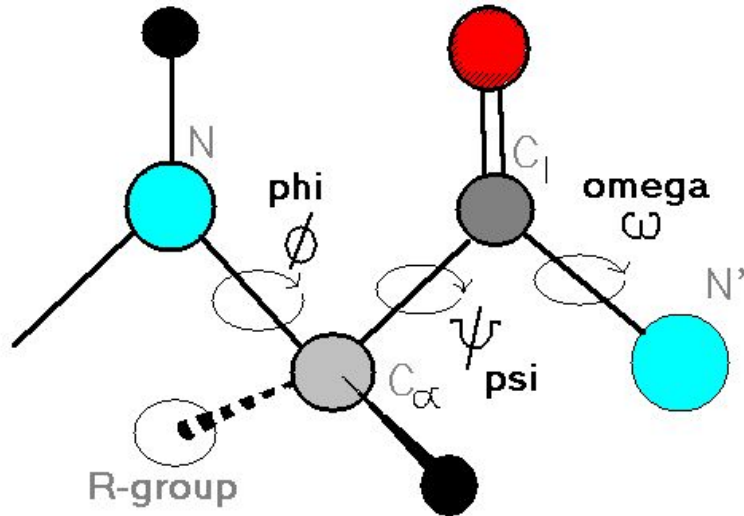
- What is light?
 - What was “the phase problem”?
 - What are the six categories of non-covalent interactions? Why do we care?
-
- Light: composed of photons, behaves like a wave and a particle, has no mass, has a distinct energy determined by its frequency, results from electromagnetic radiation
 - Phase problem: X-ray diffraction, can't know the individual phases of light
 - Six categories- can you guess?

To Knows Lecture 6

Lecture 6 Learning Outcomes

The Peptide Plane

- Draw the 6 atoms in the peptide plane.
- Apply the Boltzmann Distribution equation to calculate the probability of a conformation of known energy relative to the ground state.
- Identify bonds corresponding to backbone torsion angles phi, psi and omega



$$P = e^{-x/rt}$$

$$\uparrow E = \downarrow P$$

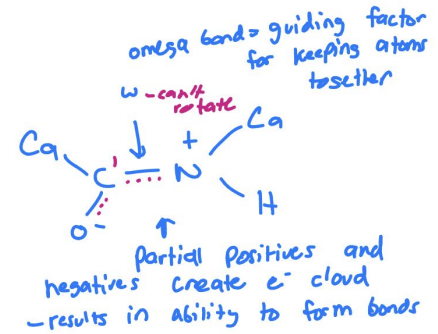
-R = gas constant

-t = temp in Kelvin

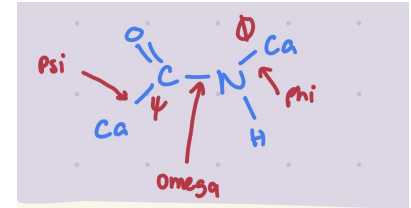
-e = energy above ground state → constant

-P = how likely it is to find a molecule with energy E

-x = energy of that state above ground state

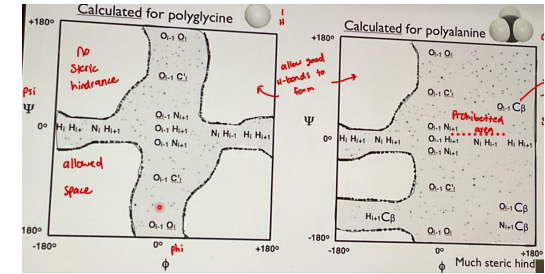


$$P = e^{-x/rt} \rightarrow \ln(P) = \frac{-x}{RT}$$



The Ramachandran Plot

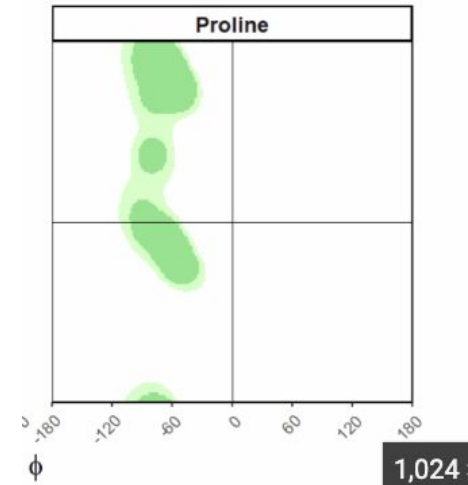
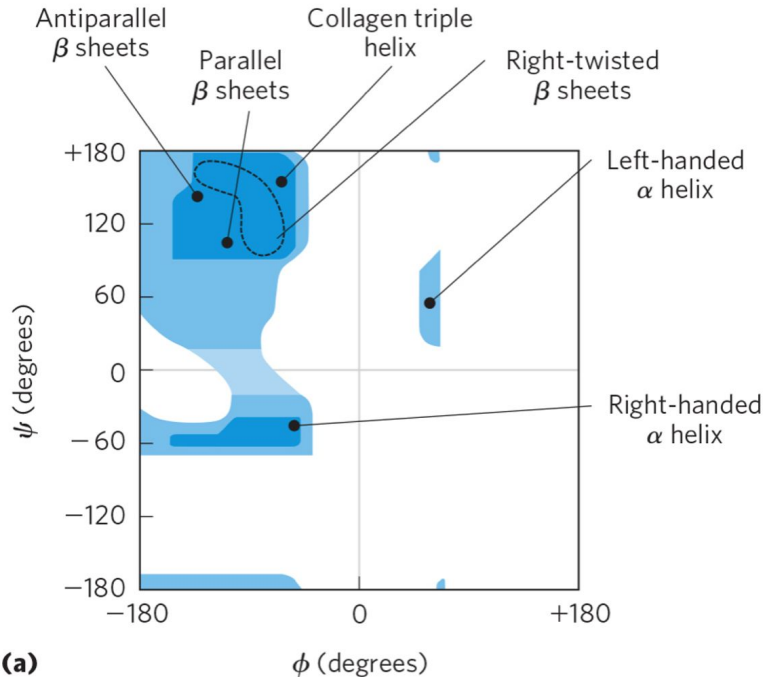
- Explain what information is displayed in the Ramachandran Plot
- Locate the allowed regions for Gly and for Pro vs. for all other AA
- Draw the general regions corresponding to alpha-helix and beta-sheet
- Understand the meaning of calculated vs. “real protein” Ramachandran Plot
- Explain why the Ramachandran Plots for folded and unfolded proteins look similar!



- Shows steric hindrance - allowed backbone confirmation. Only phi and psi values can vary
- Phi psi angles tell us the *confirmation*, get them from adjacent reactions
- Can't plot omega because it doesn't change - can't rotate
- Pro has the smallest area - least conformational freedom
- Alpha helix: psi -60, phi -40 beta sheet: psi -60, 150, phi -90, 180
- **Calculated plot:** allowed regions are boxed, map of possible conformations that only looks at generic steric hindrance - no special cases, regions instead of dots
- **Real Protein plot:** plot phi and psi for a few hundred proteins, one big map with more dots
- Because the phi psi torsion angles are adjacent to each other, whether they are folded or unfolded *adjacent interactions

The Ramachandran Plot

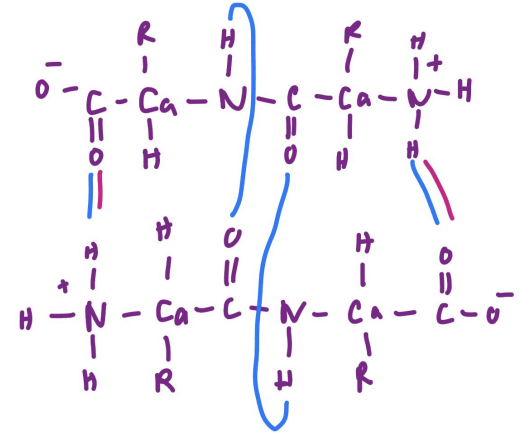
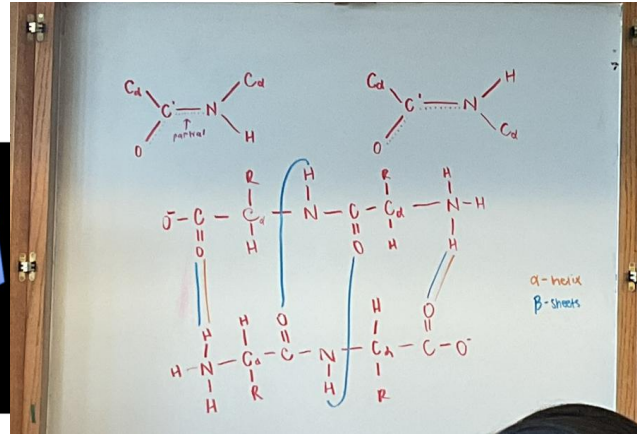
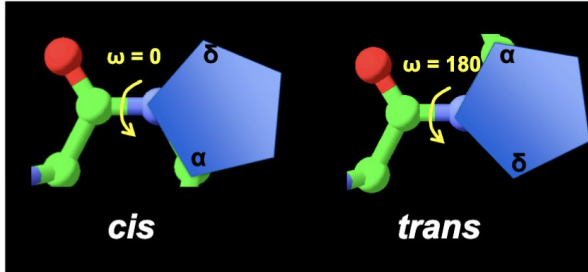
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Secondary structures

- Identify and draw cis and trans isomers of the peptide bond
- Draw and recognize the backbone hydrogen bonding and key features of the alpha-helix
- Draw and recognize the backbone hydrogen bonding and key features of the beta-sheet

cis and *trans* isomers
of the X-Pro peptide bond



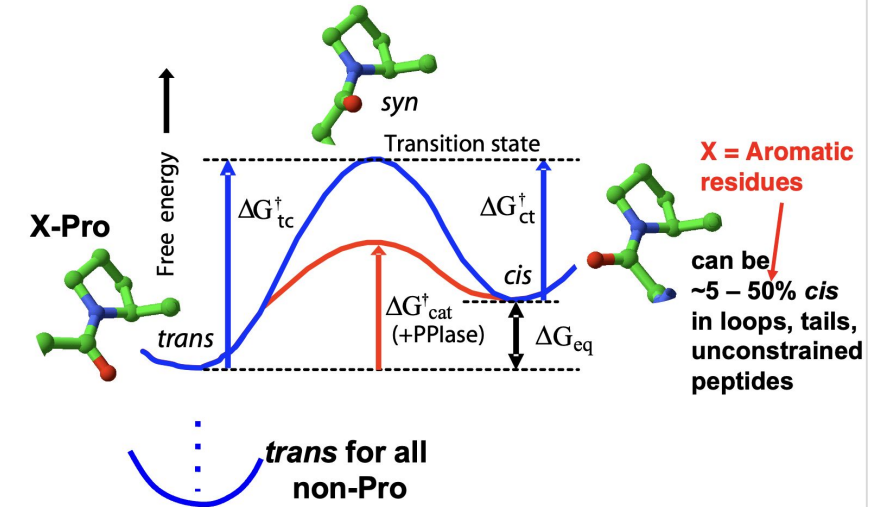
- Which one is cis? Which one is trans?

Secondary structures

- Identify and draw cis and trans isomers of the peptide bond
- Draw and recognize the backbone hydrogen bonding and key features of the alpha-helix
- Draw and recognize the backbone hydrogen bonding and key features of the beta-sheet

Pro allows population of *cis*

- Proline can exist in both cis and trans forms, while most other AA exist mainly in the trans form
- Cis creates steric clashes most of the time
- Proline has a specific ring structure -> creates steric clashing regardless of cis or trans state, so cis becomes more common
- The blue line shows it's difficult to go from trans to cis
- Peptidyl prolyl isomerase helps lower activation barrier, makes it easier to go from cis to trans



peptidyl prolyl isomerase enzymes accelerate cis-trans isomerization

To Knows Lecture 7

Lecture 7 Learning Outcomes:

Beta turns

- Describe how beta turns are defined (4 amino acid residues, $i+1$ and $i+2$ phi,psi torsion angles)
 - Identify beta turn positions where Pro is favored or where Gly is required
-
- 4 amino acid residues required for chain to turn 180 degrees
 - B-turns do NOT have repeating phi psi values ($i+1$ doesn't equal $i+2$)
 - Turns happen at the protein's surface, 7 angstroms between $Ca(i)$ and $Ca(i+3)$
 - Enables AA chain to turn, but will NEVER happen at omega angle

Pro is favoured = (i+1) position in Type I and Type II - more rigid

Gly is required = (i+2) ONLY in Type II - more flexible

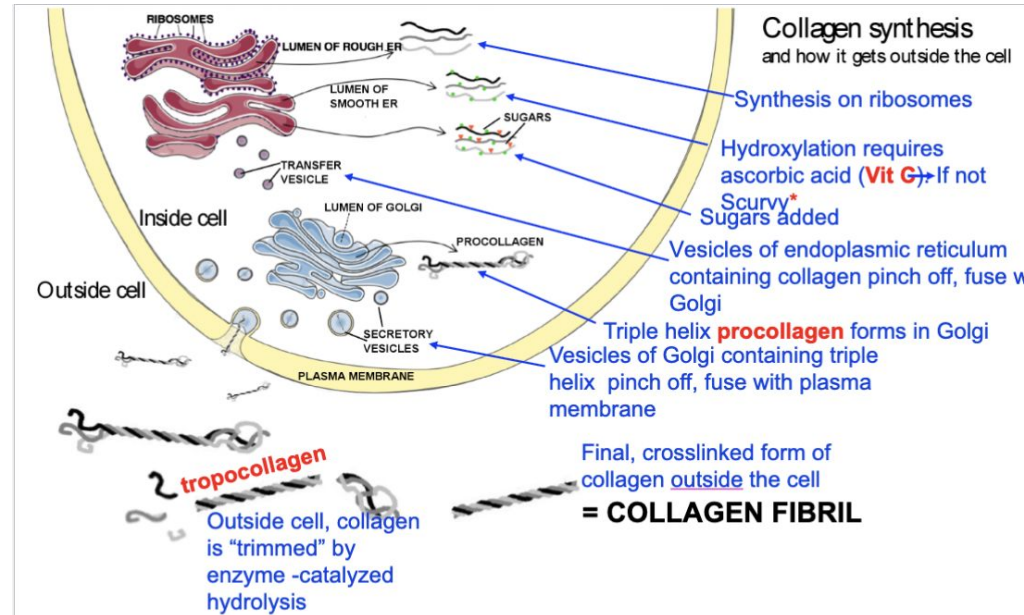
	<u>type I turn:</u>	<u>type II:</u>
i+1	proline	proline
i+2	no proline	must be glycine

could be proline



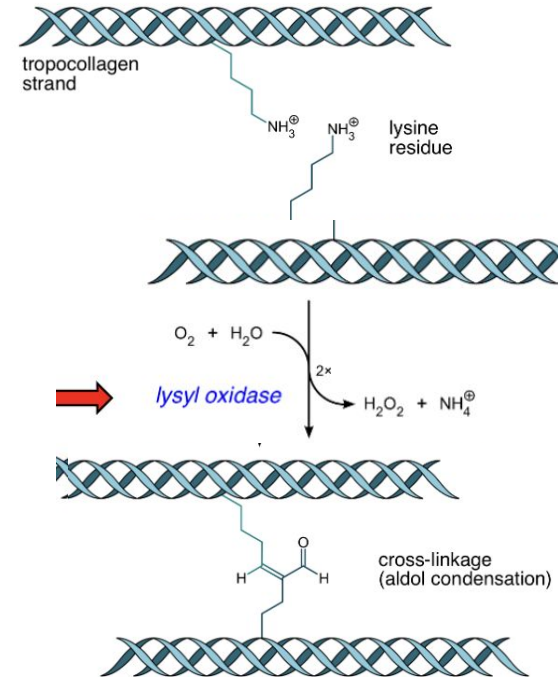
Collagen

- Describe the structure of Collagen (primary, secondary, tertiary, quaternary)
 - Describe collagen synthesis (what occurs inside vs. outside the cell)
 - Explain link between Vitamin C deficiency and Scurvy
 - Identify what amino acid sidechains undergo post-translational modification (hydroxylation, crosslinking) and where (inside vs. outside the cell) these modifications happen.
-
- “Special” secondary structure
 - (Gly-Pro-Pro) repeated, so Gly repeats every 3rd amino acid
 - Alpha chain (tertiary structure)- one long polypeptide chain of collagen
 - 3 chains come together = triple helix
 - Triple helix -> tropocollagen
 - Crosslinks: form of collagen *outside* the cell
 - Not enough vitamin C from fruits and veggies -> scurvy
 - Lysine, proline = undergoes hydroxylation
 - Lysine = only one crosslinked (outside)



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- Single-stranded triple helix -> tropocollagen
 - Crosslinks: form of collagen *outside* the cell
 - Not enough vitamin C from fruits and veggies -> scurvy
 - Lysine, proline = undergoes hydroxylation
 - Lysine = only one crosslinked (outside)
 - **Cross Linking:**
 - starts with **triple helix tropocollagen** strand with **lysine residues** on the outside
 - an enzyme called **lysyl oxidase** converts them into aldehyde carbonyl groups
 - which eventually connect in a reaction called **aldol condensation**

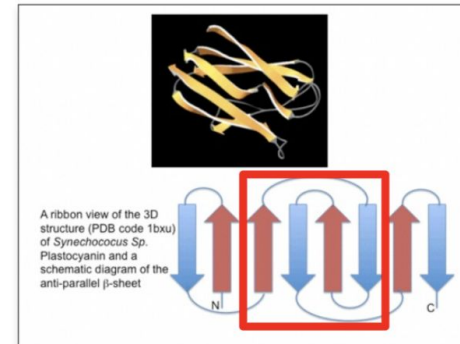


Protein Folds

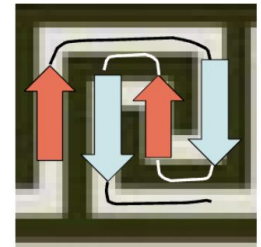
- List the main categories of folds (e.g., all alpha, all beta, etc.)
 - Give reasons for the limited number of unique protein folds.
 - Describe what a topology diagram shows
-
- All alpha, all beta, mixed alpha and beta
 - Limited ways to get stable combinations of helices, B-sheets, turns, coils
 - They can fold quickly, from milisec -> sec -> min
 - Intrinsically disordered proteins -> if polar/nonpolar, want to stay in water instead of fold
 - Peptide planes and side chains alternate orientations in b-strands, but not a-helices
 - Thermodynamics, evolutionary pressure, electron cloud constraints, backbone constraint, side chain polarity, omega bond constraints

Fold Patterns Drawn in 2D

- Topology diagram shows beta-sheet connections



Greek Key Motif (Jane Richardson, Duke U.)



To Knows Lecture 8

Lecture 8 Learning Outcomes:

- Define fold, superfamily, family
- Describe the purpose of SCOP/SCOPe, what they do and what they provide

Fold: also known as structural domain, folding unit, motif. A characteristic arrangement of the major secondary structures, including the same topological connections.

Superfamily: proteins with same fold; vary in AA sequence identity from high to nearly zero. Proteins in the same superfamily have structural (and sometimes functional) similarities that suggest a common evolutionary origin. A superfamily can contain only one, or many different families.

Family: groupings within a superfamily. These proteins have a common evolutionary origin based on either of two criteria: (1) sequence identities of > 30%; or (2) lower sequence identity but having similar structures and functions, for example: globins with sequence identities of ~ 15%.

SCOP = Structural Classification of Proteins

<http://scop.mrc-lmb.cam.ac.uk>

SCOPe = Structural Classification of Proteins extended

<http://scop.berkeley.edu>

Purpose: The purpose of both SCOP and SCOPe is to “provide a detailed and comprehensive description of the structural and evolutionary relationships between all proteins whose structure is known.”

What they do: look for structural and sequence similarities, classify protein folds (fold, family, superfamily)

How they differ: how they utilize **manual curation** vs. **automated methods** (but they agree fairly well)

What they provide: At these websites, you can find an example of every protein fold yet discovered, at least up to the last update!

KAHOOT!

<https://create.kahoot.it/share/biochem-quiz-2-kahoot/f0c4220c-4be2-45e0-a363-025994f26474>